

stats

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Packages

```
library(tidyr)
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
library(gtsummary)
library(dplyr)
library(rstatix)
library(ggpp)
library(lmerTest)
```

Data

```
# Data Frame with ALL sites, TC (utc and tc isco's), CC, IT
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
```

```

    mutate(date = mdy(date)) # telling r the date format

# Trail Creek site order
site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

# Trail Creek data
trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

# Coal Creek data
coal_creek <- compiled_isotope |>
  filter(location %in% c("cc"))

# Italian Creek data
italian_creek <- compiled_isotope |>
  filter(location == "it")

# Trail Creek ISCO data combined
trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_stddev,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

# Upstream Trail Creek ISCO
utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

# Downstream Trail Creek ISCO
tc_isco <- trail_creek_isco |>
  filter(site == "tc_isco")

# Trail Creek gradient data
gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

# Coal Creek gradient data
gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

```

```

# Italian Creek gradient data
gradient_it <- read_csv(here("data", "it_gradient_data.csv"))

# ALL Discharge data
discharge_isco <- read_csv(here("data", "isco_discharge_lc.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    period = if_else(date < as.Date("2026-05-14"), "pre-peak", "post-peak") # before & after
  )

# Upstream Trail Creek ISCO
discharge_utc_isco <- discharge_isco |>
  filter(isco %in% c("utc"))

# Downstream Trail Creek ISCO
discharge_tc_isco <- discharge_isco |>
  filter(isco %in% c("tc"))

# Upstream and Downstream Trail Creek ISCO
us_ds <- read_csv(here("data", "us_ds.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    creek = case_when(
      grepl("^cc", location) ~ "Coal Creek",
      grepl("^it", location) ~ "Italian Creek",
      location %in% c("utc", "tc") ~ "Trail Creek"),
    reach = case_when(
      location %in% c("cc-us", "it-us", "utc") ~ "upstream",
      location %in% c("cc-ds", "it-ds", "tc") ~ "downstream"))

# Summary stats per period (per reach, if you have upstream/downstream)
discharge_stat <- discharge_isco |>
  group_by(period) |> # add reach here: group_by(reach, period)
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE)
  )

```

Statistical Tests

```
# ANOVA,  
# lc-excess and location  
anova <- aov(lc_excess ~ location, data = compiled_isotope)  
summary(anova)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)  
location      3  209.2   69.72   36.53 <2e-16 ***  
Residuals    111  211.8    1.91  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# TukeyHSD, compares every possible pair of group means  
TukeyHSD(anova)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = lc_excess ~ location, data = compiled_isotope)
```

```
$location  
      diff      lwr      upr      p adj  
it-cc  2.2952900  0.8997570  3.6908230 0.0002219  
tc-cc -1.8984083 -3.0613525 -0.7354642 0.0002508  
utc-cc -0.7528817 -1.9388567  0.4330933 0.3520168  
tc-it  -4.1936983 -5.2595543 -3.1278424 0.0000000  
utc-it -3.0481717 -4.1391100 -1.9572333 0.0000000  
utc-tc  1.1455267  0.3741168  1.9169366 0.0010318
```

```
# Trail Creek  
# lc-excess & normalized stream distance  
  
# 6/24 - lc-excess & normalized stream distance  
model.1 <- lm(`lc-excess_6-24` ~ normalized_stream_distance_m, data = gradient_tc)  
summary(model.1)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

	1	2	3	4	5	6	7
	-0.36861	-0.32358	0.58444	0.03641	0.15093	0.31789	-0.39748

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.3907116	0.2562773	-5.427	0.002880 **
normalized_stream_distance_m	-0.0011403	0.0001537	-7.417	0.000702 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.416 on 5 degrees of freedom

Multiple R-squared: 0.9167, Adjusted R-squared: 0.9

F-statistic: 55.01 on 1 and 5 DF, p-value: 0.0007016

```
# 6/30 - lc-excess & normalized stream distance
```

```
model.2 <- lm(`lc-excess_6-30` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.2)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

	1	2	3	4	5	6	7
	-0.08528	-0.20669	0.24371	0.25137	0.19568	-0.61753	0.21874

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.2800413	0.2204137	-10.34	0.000145 ***
normalized_stream_distance_m	-0.0008555	0.0001322	-6.47	0.001314 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3578 on 5 degrees of freedom

Multiple R-squared: 0.8933, Adjusted R-squared: 0.872

F-statistic: 41.86 on 1 and 5 DF, p-value: 0.001314

```
# 7/15 - lc-excess & normalized stream distance
model.3 <- lm(`lc-excess_7-15` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.3)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

```
      1      2      3      4      5      6      7
-0.36655 -0.34573  0.61325  0.47415 -0.07123 -0.28328 -0.02060
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.1034551	0.2671852	-4.130	0.00909 **
normalized_stream_distance_m	-0.0010277	0.0001603	-6.412	0.00137 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4337 on 5 degrees of freedom

Multiple R-squared: 0.8916, Adjusted R-squared: 0.8699

F-statistic: 41.11 on 1 and 5 DF, p-value: 0.001369

```
# Coal Creek
# lc-excess & normalized stream distance

# 6/23 - lc-excess & normalized stream distance
model.4 <- lm(`lc-excess_6-23` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.4)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

```
      1      2      3      4      5
-0.3323  0.6279  0.2566 -0.3412 -0.2109
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.5393266	0.4492095	3.427	0.0416 *
normalized_stream_distance_m	-0.0006445	0.0002206	-2.921	0.0614 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4938 on 3 degrees of freedom

Multiple R-squared: 0.7399, Adjusted R-squared: 0.6532

F-statistic: 8.533 on 1 and 3 DF, p-value: 0.06144

```
# 7/7 - lc-excess & normalized stream distance
model.5 <- lm(`lc-excess_7-7` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.5)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

1	2	3	4	5
0.2092	0.3569	-0.8765	-0.8080	1.1184

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.333192	0.885667	-0.376	0.7318
normalized_stream_distance_m	-0.001031	0.000435	-2.369	0.0986 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9735 on 3 degrees of freedom

Multiple R-squared: 0.6517, Adjusted R-squared: 0.5356

F-statistic: 5.613 on 1 and 3 DF, p-value: 0.09859

```
# Italian Creek
# lc-excess & normalized stream distance

# 7/1 - lc-excess & normalized stream distance

model.6 <- lm(`lc_excess_7_1` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.6)
```

Call:

```
lm(formula = lc_excess_7_1 ~ normalized_stream_distance_m, data = gradient_it)
```

Residuals:

1	2	3	4	5
0.190779	-0.256938	-0.069233	0.141983	-0.006591

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.549e+00	1.634e-01	21.715	0.000214 ***
normalized_stream_distance_m	-9.279e-05	8.643e-05	-1.074	0.361692

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2061 on 3 degrees of freedom

Multiple R-squared: 0.2775, Adjusted R-squared: 0.03673

F-statistic: 1.153 on 1 and 3 DF, p-value: 0.3617

```
# 7/16 - lc-excess & normalized stream distance
```

```
model.7 <- lm(`lc-excess_7_16` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.7)
```

Call:

```
lm(formula = `lc-excess_7_16` ~ normalized_stream_distance_m,
    data = gradient_it)
```

Residuals:

1	2	3	4	5
0.19695	-0.22076	-0.11740	0.08273	0.05848

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.438e+00	1.529e-01	9.403	0.00255 **
normalized_stream_distance_m	-3.170e-04	8.088e-05	-3.919	0.02954 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1928 on 3 degrees of freedom

Multiple R-squared: 0.8366, Adjusted R-squared: 0.7822

F-statistic: 15.36 on 1 and 3 DF, p-value: 0.02954

```
# 7/23 - lc-excess & normalized stream distance
```

```
model.8 <- lm(`lc_excess_7_23` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.8)
```

Call:

```
lm(formula = lc_excess_7_23 ~ normalized_stream_distance_m, data = gradient_it)
```

Residuals:

1	2	3	4	5
0.18103	-0.20000	-0.05687	-0.04930	0.12514

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.005e+00	1.405e-01	7.154	0.00563 **
normalized_stream_distance_m	-2.609e-04	7.428e-05	-3.512	0.03914 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1771 on 3 degrees of freedom

Multiple R-squared: 0.8044, Adjusted R-squared: 0.7391

F-statistic: 12.33 on 1 and 3 DF, p-value: 0.03914

```
# Trail Creek
```

```
# lc-excess & cumulative dams
```

```
gradient_tc$cumulative_dams <- as.numeric(as.character(gradient_tc$cumulative_dams))
```

```
# 6/24 - lc-excess & cumulative dams
```

```
model.9 <- lm(`lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
summary(model.9)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

2	3	4	5	6	7
---	---	---	---	---	---

-0.4806 0.1654 0.1151 0.4332 0.4826 -0.7157

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.613391	0.362589	-4.450	0.01125 *
cumulative_dams	-0.034669	0.006788	-5.107	0.00695 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5487 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.867, Adjusted R-squared: 0.8338

F-statistic: 26.08 on 1 and 4 DF, p-value: 0.006948

```
# 6/30 - lc-excess & cumulative dams
model.10 <- lm(`lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
summary(model.10)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.27039	-0.01815	0.34655	0.43245	-0.47798	-0.01249

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.501209	0.257870	-9.699	0.000632 ***
cumulative_dams	-0.025481	0.004828	-5.278	0.006179 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3902 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8744, Adjusted R-squared: 0.843

F-statistic: 27.86 on 1 and 4 DF, p-value: 0.006179

```
# 7/15 - lc-excess & cumulative dams
model.11 <- lm(`lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
summary(model.11)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.55012	0.17680	0.52850	0.19513	-0.09976	-0.25056

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.241283	0.281127	-4.415	0.01155 *
cumulative_dams	-0.032606	0.005263	-6.195	0.00345 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4254 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9056, Adjusted R-squared: 0.882

F-statistic: 38.38 on 1 and 4 DF, p-value: 0.003452

```
# Coal Creek
# lc-excess & cumulative dams

gradient_cc$cumulative_dams <- as.numeric(as.character(gradient_cc$cumulative_dams))

# 6/23 - lc-excess & cumulative dams
model.12 <- lm(`lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
summary(model.12)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
```

Residuals:

	2	3	4	5
	0.11851	0.07466	-0.40553	0.21236

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.69377	0.41827	4.049	0.0559 .
cumulative_dams	-0.04687	0.01195	-3.921	0.0593 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3385 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8849, Adjusted R-squared: 0.8273

F-statistic: 15.37 on 1 and 2 DF, p-value: 0.05931

```
# 7/7 - lc-excess & cumulative dams
```

```
model.13 <- lm(`lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
summary(model.13)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
```

Residuals:

```
      2      3      4      5
0.8361 -0.6905 -0.9591  0.8135
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.13831	1.45089	-1.474	0.278
cumulative_dams	-0.01660	0.04146	-0.400	0.728

Residual standard error: 1.174 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.0742, Adjusted R-squared: -0.3887

F-statistic: 0.1603 on 1 and 2 DF, p-value: 0.7276

```
# Trail Creek
```

```
# lc-excess & cumulative dam length
```

```
gradient_tc$cumulative_dam_length <- as.numeric(as.character(gradient_tc$cumulative_dam_length))
```

```
# 6/24 - lc-excess & cumulative dam length
```

```
model.14 <- lm(`lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
summary(model.14)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.3782	0.3121	-0.2486	0.3477	0.3552	-0.3882

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.715833	0.259857	-6.603	0.00273 **
cumulative_dam_length	-0.056153	0.008125	-6.911	0.00230 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4183 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9227, Adjusted R-squared: 0.9034

F-statistic: 47.77 on 1 and 4 DF, p-value: 0.002299

```
# 6/30 - lc-excess & cumulative dam length
model.15 <- lm(`lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
summary(model.15)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.16963	0.11233	0.09129	0.36273	-0.58878	0.19207

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.601968	0.233442	-11.15	0.000369 ***
cumulative_dam_length	-0.040214	0.007299	-5.51	0.005295 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3758 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8836, Adjusted R-squared: 0.8545

F-statistic: 30.36 on 1 and 4 DF, p-value: 0.005295

```
# 7/15 - lc-excess & cumulative dam length
model.16 <- lm(`lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
summary(model.16)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

```
      2      3      4      5      6      7
-0.4429  0.3245  0.1916  0.1118 -0.2269  0.0420
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.348532	0.197391	-6.832	0.00240 **
cumulative_dam_length	-0.052360	0.006172	-8.484	0.00106 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3178 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9474, Adjusted R-squared: 0.9342

F-statistic: 71.97 on 1 and 4 DF, p-value: 0.001058

```
# Coal Creek
# lc-excess & cumulative dam length

gradient_cc$cumulative_dam_length <- as.numeric(as.character(gradient_cc$cumulative_dam_length))

# 6/23 - lc-excess & cumulative dam length
model.17 <- lm(`lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
summary(model.17)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

```
      2      3      4      5
0.10405 -0.06714 -0.20308  0.16617
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.547568	0.225772	6.855	0.0206 *
cumulative_dam_length	-0.027556	0.004094	-6.730	0.0214 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2052 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9577, Adjusted R-squared: 0.9366

F-statistic: 45.29 on 1 and 2 DF, p-value: 0.02137

```
# 7/7 - lc-excess & cumulative dam length
model.18 <- lm(`lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
summary(model.18)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

2	3	4	5
0.7792	-0.7622	-0.8651	0.8481

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.11270	1.26808	-1.666	0.238
cumulative_dam_length	-0.01134	0.02300	-0.493	0.671

Residual standard error: 1.152 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.1083, Adjusted R-squared: -0.3375

F-statistic: 0.243 on 1 and 2 DF, p-value: 0.6709

```
# Trail Creek ISCO's
# Correlation test of discharge and lc-excess

# Downstream Trail Creek ISCO
cor.test(discharge_tc_isco$discharge, discharge_tc_isco$lc_excess)
```

Pearson's product-moment correlation

```
data: discharge_tc_isco$discharge and discharge_tc_isco$lc_excess
t = 24.555, df = 44, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9379629 0.9808167
sample estimates:
      cor
0.9653947
```

```
# Upstream Trail Creek ISCO
cor.test(discharge_utc_isco$discharge, discharge_utc_isco$lc_excess)
```

Pearson's product-moment correlation

```
data: discharge_utc_isco$discharge and discharge_utc_isco$lc_excess
t = 4.7379, df = 38, p-value = 3e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3676755 0.7739744
sample estimates:
      cor
0.6093889
```

```
# Downstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_tc_isco)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)
```

Coefficients:

(Intercept)	discharge
-5.986	42.209

```
# Upstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_utc_isco)
```

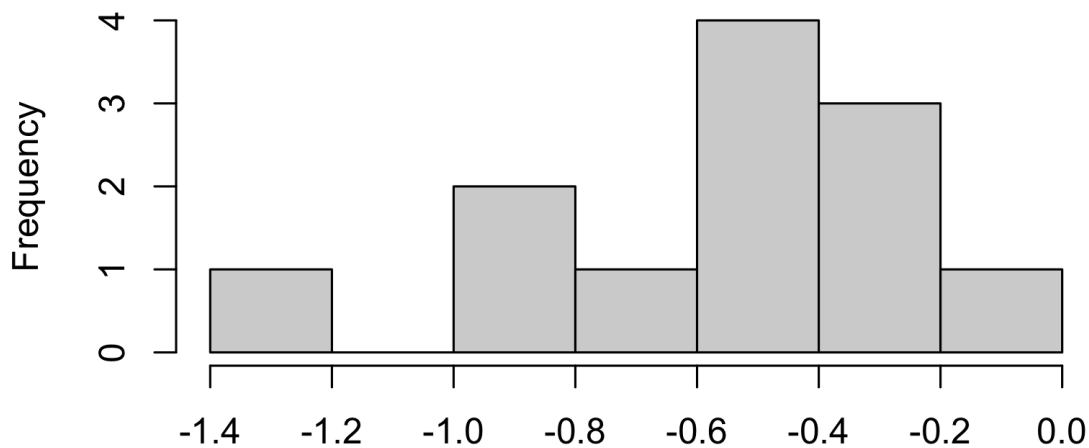


```
Call:
lm(formula = lc_excess ~ discharge, data = discharge_utc_isco)
```

```
Coefficients:
(Intercept)    discharge
      -2.975         19.597
```

```
# Visual check
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]) # skewed left
```

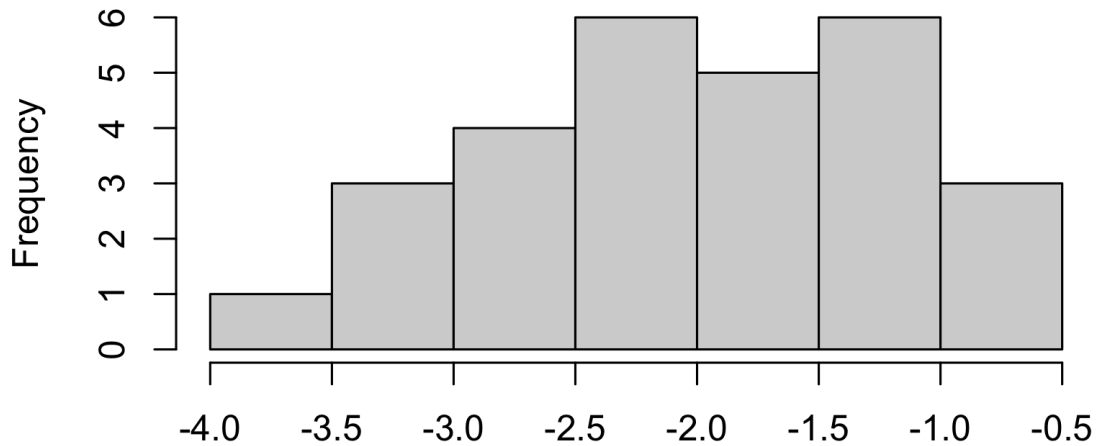
f discharge_utc_isco\$lc_excess[discharge_utc_isco\$period == "pre-peak"]



discharge_utc_isco\$lc_excess[discharge_utc_isco\$period == "pre-peak"]

```
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]) # normal distribution
```

discharge_utc_isco\$lc_excess[discharge_utc_isco\$period



discharge_utc_isco\$lc_excess[discharge_utc_isco\$period == "post-pea

```
# Formal test (Shapiro-Wilk) - only reliable with reasonable sample size  
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"])
```

Shapiro-Wilk normality test

```
data: discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]  
W = 0.97691, p-value = 0.9683
```

```
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"])
```

Shapiro-Wilk normality test

```
data: discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]  
W = 0.96427, p-value = 0.4377
```

```
# Parametric alternative (Welch's t-test, doesn't assume equal variance) - use if normal
t.test(lc_excess ~ period, data = discharge_utc_isco)
```

Welch Two Sample t-test

```
data:  lc_excess by period
t = -8.1949, df = 37.963, p-value = 6.38e-10
alternative hypothesis: true difference in means between group post-peak and group pre-
peak is not equal to 0
95 percent confidence interval:
 -1.784032 -1.077199
sample estimates:
mean in group post-peak  mean in group pre-peak
      -1.9979821          -0.5673667
```

```
discharge_utc_isco |>
  group_by(period) |>
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE)
  )
```

```
# A tibble: 2 x 5
  period      n mean_lc median_lc sd_lc
  <chr>    <int>   <dbl>    <dbl> <dbl>
1 post-peak    28   -2.00    -2.02  0.771
2 pre-peak     12   -0.567   -0.542  0.333
```

```
var.test(lc_excess ~ period, data = discharge_utc_isco)
```

F test to compare two variances

```
data:  lc_excess by period
F = 5.3505, num df = 27, denom df = 11, p-value = 0.005712
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
```

```

1.702796 13.452718
sample estimates:
ratio of variances
      5.350494

```

```

# Correlation of the lc-excess & location

compiled_isotope$date_c <- as.numeric(compiled_isotope$date) - mean(as.numeric(compiled_isotope$date))
# center the date variable by subtracting its mean.

model.lmer <- lmer(
  lc_excess ~ location + date_c + # fixed effects: location differences + linear time trend
  (1|site), # random intercept: allows each site to have its own baseline lc-excess, accounting for
  # the ~36% of variance that's due to site-level clustering rather than location differences
  data = compiled_isotope)

summary(model.lmer)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: lc_excess ~ location + date_c + (1 | site)
Data: compiled_isotope

```

REML criterion at convergence: 334.7

```

Scaled residuals:
      Min       1Q   Median       3Q      Max
-2.18174 -0.59314 -0.05398  0.65138  2.16327

```

```

Random effects:
Groups   Name              Variance Std.Dev.
site     (Intercept)  0.4233    0.6506
Residual                    0.8648    0.9300
Number of obs: 115, groups: site, 20

```

```

Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)  0.28574    0.38862   25.19091   0.735    0.4690
locationit    2.91690    0.53729   19.55019   5.429 2.79e-05 ***
locationtc   -2.88718    0.52093   16.75436  -5.542 3.77e-05 ***
locationutc  -1.55173    0.54866   15.90779  -2.828  0.0122 *
date_c       -0.04662    0.00458  109.92129 -10.179 < 2e-16 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
(Intr) loctnt locationtc locationutc
locationit -0.656
locationtc -0.722 0.501
locationutc -0.692 0.473 0.506
date_c      -0.236 -0.114 0.073 0.099
```

```
# centering doesn't affect slopes, only the intercept's ^ from the (1|site)
```

```
anova(model.lmer)
```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
location	111.828	37.276	3	15.082	43.102	1.238e-07 ***
date_c	89.602	89.602	1	109.921	103.605	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# the enrichment effect consistent across geologically distinct sites? This is a reach × creek interaction
```

```
model <- lmer(lc_excess ~ reach * creek + (1 | date), data = us_ds)
summary(model)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

lmerModLmerTest]

Formula: lc_excess ~ reach * creek + (1 | date)

Data: us_ds

REML criterion at convergence: 27.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.05323	-0.31166	-0.03435	0.35767	0.96783

Random effects:

Groups	Name	Variance	Std.Dev.
date	(Intercept)	1.19749	1.0943
Residual		0.09793	0.3129

Number of obs: 16, groups: date, 8

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-1.3596	0.8048	5.3922	-1.689	0.14768
reachupstream	1.9010	0.3129	5.0000	6.075	0.00175
creekItalian Creek	2.7533	1.0390	5.3922	2.650	0.04217
creekTrail Creek	-3.3642	1.0390	5.3922	-3.238	0.02064
reachupstream:creekItalian Creek	-1.1078	0.4040	5.0000	-2.742	0.04069
reachupstream:creekTrail Creek	1.0219	0.4040	5.0000	2.530	0.05256

(Intercept)
reachupstream **
creekItalian Creek *
creekTrail Creek *
reachupstream:creekItalian Creek *
reachupstream:creekTrail Creek .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	rchpst	crkItC	crkTrC	rch:IC
reachupstrm	-0.194				
crekItlnCrk	-0.775	0.151			
crekTrlCrk	-0.775	0.151	0.600		
rchpstrm:IC	0.151	-0.775	-0.194	-0.117	
rchpstrm:TC	0.151	-0.775	-0.117	-0.194	0.600

`anova(model)`

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
reach	13.5224	13.5224	1	5	138.082	7.849e-05 ***
creek	3.0212	1.5106	2	5	15.425	0.007264 **
reach:creek	3.4032	1.7016	2	5	17.376	0.005611 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

the lmer() model is more powerful here because it pools information across all three creeks

What the model is actually telling you:

```
# `reach` main effect (p = 7.8e-05): across all sites, upstream vs. downstream location matters
# `creek` main effect (p = 0.0073): the three creeks differ in baseline lc-excess overall, with Italian Creek having the highest
# `reach:creek` interaction (p = 0.0056): this is the key term for aim (ii). It's significant!
# Looking at the coefficients:
# Italian Creek's interaction term is -1.11 (p = 0.041) - the upstream/downstream gap is smaller there
# Italian Creek is your undammed control, so a weaker reach effect there strengthens your case for the dam
# Trail Creek's interaction term is +1.02 (p = 0.053, right at the edge) - trending toward a larger gap
```